

PCA Step-by-Step Example

Complete Mathematical Walkthrough with Theory

What is Principal Component Analysis (PCA)?

PCA finds orthogonal directions (principal components) that capture maximum variance in the data. It transforms correlated variables into uncorrelated components.

Goal: Reduce dimensionality while preserving as much information (variance) as possible

Mathematical Foundation

For data matrix X ($m \times n$): m = observations, n = features

$$\text{Mean: } \mu_j = (1/m) \sum_{i=1}^m x_{ij}$$

$$\text{Centered data: } \tilde{X} = X - \mu$$

$$\text{Covariance matrix: } C = (1/(m-1)) \tilde{X}^T \tilde{X}$$

$$\text{Eigenvalue equation: } C v = \lambda v$$

Eigenvectors (v) = directions of maximum variance

Eigenvalues (λ) = amount of variance in each direction

Given Data:

F	Ex1	Ex2	Ex3	Ex4
X_1	4	8	13	7
X_2	11	4	5	14

Step 1: Calculate Mean

Why? We need to center the data around zero so PCA finds directions of variation relative to the mean, not the origin.

$$\mu_j = (1/m) \sum_{i=1}^m x_{ij}$$

$$\bar{X}_1 = \frac{1}{2}(4 + 8 + 13 + 7) = \frac{1}{2}(32) = 8$$

$$\bar{X}_2 = \frac{1}{4}(11 + 4 + 5 + 14) = \frac{1}{4}(34) = 8.5$$

✓ Mean vector $\mu = [8, 8.5]^T$

Step 2: Center the Data

Subtract the mean from each data point to get centered data $\tilde{X}$

$$\tilde{X} = X - \mu$$

Centered data:

$$\text{Ex1: } [4-8, 11-8.5] = [-4, 2.5]$$

$$\text{Ex2: } [8-8, 4-8.5] = [0, -4.5]$$

$$\text{Ex3: } [13-8, 5-8.5] = [5, -3.5]$$

$$\text{Ex4: } [7-8, 14-8.5] = [-1, 5.5]$$

Step 3: Calculate Covariance Matrix

The covariance matrix captures how features vary together. Diagonal = variance, off-diagonal = covariance

$$C = (1/(n-1)) \tilde{X}^T \tilde{X}$$

$$C = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$$

Calculating $\text{Cov}(X_1, X_2)$:

$$\begin{aligned} \text{Cov}(X_1, X_2) &= (1/(n-1)) \sum_{i=1}^n (x_{1i} - \mu_1)(x_{2i} - \mu_2) \\ &= \frac{1}{4}[(-4)(2.5) + (0)(-4.5) + (5)(-3.5) + (-1)(5.5)] \\ &= \frac{1}{4}[-10 + 0 - 17.5 - 5.5] = \frac{1}{4}(-33) = -8.25 \end{aligned}$$

Calculating $\text{Cov}(X_1, X_1) = \text{Var}(X_1)$:

$$= \frac{1}{4}[(-4)^2 + (0)^2 + (5)^2 + (-1)^2] = \frac{1}{4}[16 + 0 + 25 + 1] = 8.25$$

Calculating $\text{Cov}(X_2, X_2) = \text{Var}(X_2)$:

$$= \frac{1}{4}[(2.5)^2 + (-4.5)^2 + (-3.5)^2 + (5.5)^2] = \frac{1}{4}[6.25 + 20.25 + 12.25 + 30.25] = 17.25$$

$$C = \begin{bmatrix} 8.25 & -8.25 \\ -8.25 & 17.25 \end{bmatrix}$$

✓ Covariance matrix is symmetric: $\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1)$

Step 4: Find Eigenvalues

Eigenvalues tell us how much variance each principal component captures. Larger eigenvalue = more important direction.

Characteristic equation: $\det(C - \lambda I) = 0$

$$\begin{aligned}\det\begin{pmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{pmatrix} &= 0 \\ &= (14-\lambda)(23-\lambda) - (-11)(-11) \\ &= 322 - 14\lambda - 23\lambda + \lambda^2 - 121 \\ &= \lambda^2 - 37\lambda + 201 = 0\end{aligned}$$

Using quadratic formula: $\lambda = (37 \pm \sqrt{37^2 - 4 \times 1 \times 201})/2$

$$\lambda = (37 \pm \sqrt{1369 - 804})/2 = (37 \pm \sqrt{565})/2$$

$$\lambda_1 = 30.3849, \lambda_2 = 6.6151$$

$$\checkmark \lambda_1 + \lambda_2 = 37 = \text{trace}(C) = 14 + 23 \checkmark$$

$$\checkmark \lambda_1 \times \lambda_2 = 201 = \det(C) = 14 \times 23 - (-11)^2 \checkmark$$

Step 5: Find Eigenvectors

Eigenvectors show the directions of principal components. They must be normalized (unit length).

For each λ_i , solve: $(C - \lambda_i I)v_i = 0$

For $\lambda_1 = 30.3849$:

$$\begin{aligned}\begin{pmatrix} 14-30.3849 & -11 \\ -11 & 23-30.3849 \end{pmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{pmatrix} -16.3849 & -11 \\ -11 & -7.3849 \end{pmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}\end{aligned}$$

$$\text{From first row: } -16.3849u_1 - 11u_2 = 0$$

$$\text{So: } u_2/u_1 = -16.3849/(-11) = 1.4895$$

$$\text{Let } u_1 = 1, \text{ then } u_2 = -1.4895$$

$$\text{Normalize: } ||v|| = \sqrt{1^2 + (-1.4895)^2} = \sqrt{3.2186} = 1.7940$$

$$e_1 = [1/1.7940] = [0.5574]$$

$$[-1.4895/1.7940] \quad [-0.8303]$$

For $\lambda_2 = 6.6151$:

Similarly solving $(C - \lambda_2 I)v_2 = 0$:

$$e_2 = [0.8303]$$

$$[0.5574]$$

$$\text{Check orthogonality: } e_1^T e_2 = 0.5574 \times 0.8303 + (-0.8303) \times 0.5574 = 0 \checkmark$$

Step 6: Calculate Principal Components

Project the centered data onto the eigenvector directions to get PC scores

$$PC_1 = e_1^T \times \tilde{X}$$

$$PC_2 = e_2^T \times \tilde{X}$$

For Ex1: $PC_1 = [0.5574, -0.8303] \times [-4, 2.5] = 0.5574 \times (-4) + (-0.8303) \times 2.5 = -4.305$

For Ex2: $PC_1 = [0.5574, -0.8303] \times [0, -4.5] = 0.5574 \times 0 + (-0.8303) \times (-4.5) = 3.736$

For Ex3: $PC_1 = [0.5574, -0.8303] \times [5, -3.5] = 0.5574 \times 5 + (-0.8303) \times (-3.5) = 5.693$

For Ex4: $PC_1 = [0.5574, -0.8303] \times [-1, 5.5] = 0.5574 \times (-1) + (-0.8303) \times 5.5 = -5.124$

Component	Ex1	Ex2	Ex3	Ex4
PC_1	-4.305	3.736	5.693	-5.124

Step 7: Interpret Results

Variance explained by each component:

Variance explained by $PC_1 = \lambda_1 / (\lambda_1 + \lambda_2) = 30.38 / (30.38 + 6.62) = 82.1\%$

Variance explained by $PC_2 = \lambda_2 / (\lambda_1 + \lambda_2) = 6.62 / (30.38 + 6.62) = 17.9\%$

PC_1 captures most of the variation in the data!

Step 8: Geometric Interpretation

What do the eigenvectors mean?

- $e_1 = [0.5574, -0.8303]$: Points in direction of maximum variance
- $e_2 = [0.8303, 0.5574]$: Perpendicular to e_1 , captures remaining variation
- PC scores show how far each point projects onto these directions
- Negative PC_1 scores mean data points are on opposite side of mean

Why is PCA Useful?

1. Dimensionality Reduction: $2D \rightarrow 1D$ using just PC_1 (keeps 82% of info!)
2. Data Visualization: Plot data in 2D PC space instead of original coordinates
3. Feature Extraction: PC scores become new features for machine learning
4. Noise Reduction: Keep only top PCs, discard noisy dimensions

Key Takeaway: PCA transforms correlated data into uncorrelated components ordered by importance. The first few components capture most of the variation, allowing effective dimensionality reduction while preserving information!

Quick Reference: Key Equations

Step	Formula
Mean	$\mu_j = (1/m) \sum_{i=1}^m x_{ij}$
Covariance Matrix	$C = (1/(m-1)) \tilde{X}^T \tilde{X}$
Eigenvalues	$\det(C - \lambda I) = 0$
Eigenvectors	$(C - \lambda_i I) \mathbf{v}_i = 0$, normalize $\ \mathbf{v}_i\ = 1$
PC Scores	$PC_i = \mathbf{e}_i^T \times \tilde{X}$